Practice Exam - 1

Question 16 of 117

A 36 y/o woman who delivered a child 3 weeks prior presents to the ED with rest dyspnea, abdominal pain, and edema. On examination, she is uncomfortable with HR=115, respiratory rate=18, and BP=94/70. There are rales 1/3 up the posterior lung fields. The JVP is to the angle of the jaw. There is an audible S3 and a 2/6 systolic murmur at the apex. The liver is enlarged and pulsatile, and there is pitting edema to the knees.Her extremities are cool.

A point of care echo in the ED demonstrates an EF of 15% with moderate MR. The RV is moderately dysfunctional.

She is admitted to the CCU and given a dose of furosemide without significant urine output. A pulmonary artery catheter is placed and shows the following hemodynamics:

RA=20, PA=56/32, PCWP=30 with a cardiac output and index of 2.4 l/min and 1.5 l/min/m2, respectively.

Which of the following is likely to improve the hemodynamics acutely?

A. Ivabradine

B. Sodium nitroprusside

C. Dopamine

D. Sacubitril/Valsartan

Commentary

Correct Answer: B. Sodium nitroprusside

Rationale: It should be recognized that this patient has the clinical and hemodynamic profile of a patient in cardiogenic shock from peri-partum cardiomyopathy. While she is hypotensive, she is also vasoconstricted with an SVR of 1900 dyne s/cm5 (Candidates should be comfortable calculating SVR - (MAP-CVP)/CO*80. In this case, MAP can be approximated as (DBP + 1/3PP) or 78, giving)78-

20))/2.4*80=1933 dynes*s/cm5. In this setting, placement of an arterial catheter to measure systemic blood pressure in real time and administration of a short-acting intravenous vasodilator such as nitroprusside will often result in significant improvement in the hemodynamic profile without requiring administration of inotropic therapy.

Answer A: Ivabradine slows heart rate. Although this patient is tachycardic, this is a compensatory response to her critically low stroke volume. Decreasing heart rate without improving stroke volume (which can be done through vasodilation), can be very dangerous in such a patient.

Answer C: Dopamine is a weak inotrope and vasoconstrictor. This will not improve the patient's hemodynamics, and may increase SVR further leading to a fall in CO/CI.

Answer D: Sacubitril/valsartan may have the same effect as sodium nitroprusside to reduce SVR in this patient, but is a poor choice due to the presence of shock and hemodynamic instability. Sacubitril-valsartan is longer-acting than sodium nitroprusside and if the patient does not tolerate vasodilation may destabilize her and require urgent management of worsening hypotension while waiting for the oral dose to wear off. In addition, the patient is currently anuric and hypoperfusing, putting the kidney at risk for worsening hemodynamic injury. In such cases, angiotensin blocking drugs should be avoided until the injury has improved, making sodium nitroprusside a better choice.